Task Description | End-Sem Practical Examination | EE-B | Batch 2

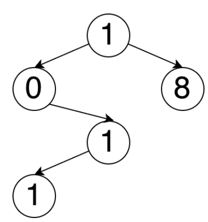
Write a C program that accepts the **last 5 digits** of your own 11-digit roll number as an array of digits. For example, if the roll number is **25295918011**, then, the last five digits are **18011** and it should be stored as {1, 8, 0, 1, 1}. The program should perform Binary Search Tree operations and traversals based on the roll number.

Tasks to be performed

1. Input the last 5 digits of the roll number into an integer array (take user input).

2. Preserve the original copy of the array for later use.

3. Insert the digits in the original order into a BST using the following rule.
    a. If the new digit is smaller than or equal to the current node, insert it in the left subtree.
    b. If the new digit is greater than the current node, insert it in the right subtree.



4. Print the following traversals of the BST:
    a. Inorder Traversal (For instance, **01118** is the inorder output for the given example)
    b. Pre–Order Traversal (For instance, **10118** is the pre-order output for the given example)

5. Perform Level Ordered Traversal on this BST using array implementation of Queue. While doing the traversal, print the level of each node as follows:

Level of Nodes 1, 2, 3, 4, 5 = L1, L2, L3, L4, L5

6. Perform Post Order Traversal of the tree. Calculate the height and depth of each node in the BST while traversing and print it as follows.

Height of Node 1 = H1, Depth of Node 1 = Y1
Height of Node 2 = H2, Depth of Node 2 = Y2
Height of Node 3 = H3, Depth of Node 3 = Y3
Height of Node 4 = H4, Depth of Node 4 = Y4
Height of Node 5 = H5, Depth of Node 5 = Y5

Important: Each print output must be on a new line. If the given example outputs do not contain whitespaces between digits, then your output must also not contain whitespaces.